MILESTONE-1: Orientation Presentation Document

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1. Introduction

The hospitality industry generates large volumes of data related to bookings, pricing, customer behaviour, and seasonal demand. Managing and analyzing this data manually is challenging and time-consuming. Infosys_HotelRevAI is an AI-driven revenue analysis approach designed to help hotels make smarter, data-driven decisions.

In this project, we focus on building Power BI dashboards to analyze hotel revenue performance using historical data and AI-based insights. The goal is to understand revenue trends, identify key factors affecting profitability, and support strategic decision-making.

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2. Problem Statement

Hotels face several challenges in revenue management, such as:

- Fluctuating demand due to seasons, holidays, and events
- Frequent booking cancellations
- Difficulty in setting optimal room prices
- Lack of real-time and visual insights for decision-making

Traditional reporting methods are not sufficient to handle complex and large datasets. Hence, an AI-driven and visualization-based solution is required.

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3. Project Objective

The main objectives of Infosys_HotelRevAI are:

- To analyze hotel revenue data using Power BI
- To identify trends in bookings, cancellations, and revenue
- To calculate key performance indicators (KPIs) such as ADR and occupancy rate
- To provide interactive dashboards for easy understanding
- To support revenue optimization decisions using data insights

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4. What is AI-Driven Revenue Analysis?

AI-driven revenue analysis uses advanced techniques such as:

- Data analytics
- Pattern recognition
- Predictive insights

In the context of hotels, AI helps in:

- Understanding customer booking behavior
- Predicting demand patterns
- Identifying high and low revenue periods
- Supporting dynamic pricing strategies

Although the dashboards are built in Power BI, AI concepts are applied through trend analysis, forecasting visuals, and data-driven insights.

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5. Dataset Overview

The hotel revenue dataset typically includes the following attributes:

- Hotel type (City Hotel / Resort Hotel)
- Booking status (Canceled / Not Canceled)
- Lead time
- Arrival date (year, month, week)
- Number of adults, children, and babies
- Room type and meal plan
- Average Daily Rate (ADR)
- Number of nights stayed

This raw dataset is cleaned and transformed using Power Query before visualization.

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6. Key Metrics Used

Some important metrics used in the analysis are:

- Revenue
  Revenue = ADR × Number of Nights Stayed
- ADR (Average Daily Rate)
  Average revenue earned per occupied room per day
- Occupancy Rate
  Percentage of rooms occupied during a given period
- Cancellation Rate
  Percentage of bookings that were canceled

These metrics help evaluate hotel performance clearly.

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7. Role of Power BI in the Project

Power BI plays a crucial role in this project by:

- Connecting to hotel revenue datasets
- Cleaning and transforming data using Power Query
- Creating interactive dashboards and reports
- Using filters, slicers, and visuals for analysis

Power BI dashboards make complex data easy to understand for both technical and non-technical users.

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8. Dashboards Created

The Power BI dashboards include:

- Revenue trend analysis (monthly and yearly)
- ADR comparison across hotel types
- Booking vs cancellation analysis
- Seasonal and demand-based insights
- Performance comparison between City Hotels and Resort Hotels

These dashboards allow users to explore data interactively.

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9. Benefits of Infosys_HotelRevAI

- Improves revenue forecasting accuracy
- Helps management make informed pricing decisions
- Identifies revenue leakages due to cancellations
- Saves time through automated analysis
- Enhances strategic planning using visual insights

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10. Conclusion

Infosys_HotelRevAI demonstrates how AI-driven analytics combined with Power BI can transform hotel revenue management. By converting raw hotel data into meaningful dashboards, hotels can better understand their performance, predict trends, and maximize revenue.

This project highlights the importance of data visualization and AI-based insights in the modern hospitality industry.

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11. Future Scope

- Integration of real-time hotel data
- Advanced AI-based demand forecasting models
- Personalized pricing strategies
- Expansion to multi-hotel or chain-level analysis